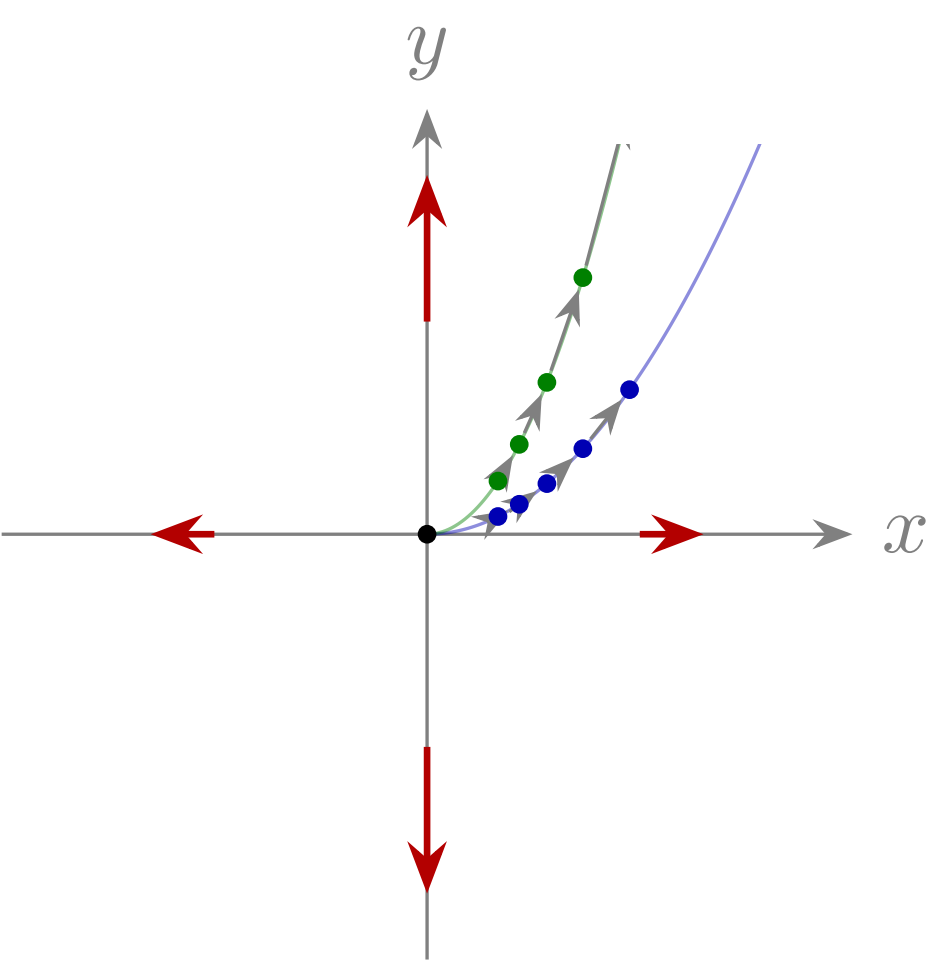

$\alpha = 1.3,\ \beta = 1.69$